

Practice Problem (Module 6)

A researcher is studying how daily screen time and the type of electronic device primarily used relate to how many hours of sleep a person gets per night. The researcher fits two linear regression models using a sample of 90 individuals:

- `sleep_hours`: average hours of sleep per night
- `screen_time`: average screen time per day in hours
- `device_type`: primary device used most during the day (categorical predictor with 3 levels: "Phone", "Tablet", "Laptop")

Call:

```
lm(formula = sleep_hours ~ screen_time + device_type, data = sleep_df)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.2000	0.3000	27.33	<2e-16 ***
screen_time	-0.4500	0.0500	-9.00	<2e-16 ***
device_typeTablet	-0.8000	0.4000	-2.00	0.048 *
device_typeLaptop	-1.5000	0.4200	-3.57	0.0006 ***

Note: For any problem that asks for null and alternative hypotheses, write them in terms of betas.

1.

a. Write the fitted model.

b. Interpret the intercept and the coefficient for `device_typeTablet`.

c. Write the null and alternative hypotheses to test whether `device_type` has an effect on `sleep_hours`. What R function can be used to test this?

d. Write the null and alternative hypotheses to test whether the average `sleep_hours` is the same for users of tablets and laptops. What R function can be used to test this?

Call:

```
lm(formula = sleep_hours ~ screen_time * device_type, data = sleep_df)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.2000	0.3300	24.85	<2e-16 ***
screen_time	-0.4500	0.0600	-7.50	<2e-16 ***
device_typeTablet	-0.8000	0.4700	-1.70	0.092
device_typeLaptop	-1.5000	0.4800	-3.13	0.0022 **
screen_time:device_typeTablet	0.1000	0.0800	1.25	0.215
screen_time:device_typeLaptop	0.3000	0.0850	3.53	0.0007 ***

2. (Use the model on the previous page)

a. Write the fitted model including the interaction terms.

b. Write the null and alternative hypotheses to test whether there is an interaction between screen time and device type. What R function can be used to test this?

c. Write the null and alternative hypotheses to test whether screen_time affects sleep_hours for **laptop users**. What R function can be used to evaluate this?

d. What is the estimated difference in average sleep_hours between **tablet users and laptop users** when screen_time = 4?

e. Interpret the coefficient for screen_time for both models (with interaction and without). What is different between the two interpretations?